CHALLA VARUN KUMAR

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SUMMARY

Dedicated Computer Science student at Kalasalingam Academy of Research and Education with a 9.37 CGPA and a specialization in AI & Machine Learning. Proven ability to develop complex, full-stack applications using React, Next.js, and Java, including a Gemini API-integrated educational platform. Skilled in modern DevOps tools like Docker and CI/CD pipelines to ensure efficient, scalable deployment of software solutions. Passionate about solving real-world problems through innovation, as evidenced by a filed patent for a Smart Irrigation System featuring AI-based decision making.

EDUCATION

Aug 2024 - Present

Bachelor of Technology in Computer Science, Kalasalingam Academy of Research and Education
B Tech - CSE - AI & ML
Current CGPA : 9.37

SKILLS

- **Programming Languages:** Java, Python, C Programming , HTML5, CSS3
- **Disruptive Technologies:** Artificial Intelligence (AI), Machine Learning (ML), Internet of Things (IoT)
- **Web Frameworks & Libraries:** React, Next.js, Node.js, Tailwind CSS
- **Database Management:** PostgreSQL, Supabase, Relational Databases, Big Data, Data Warehousing
- **Tools & DevOps:** Git , GitHub ,Docker, CI/CD Pipelines

WORK EXPERIENCE

OWASP Student Chapter – Web Development Team Member
Kalasalingam Academy of Research and Education

- Optimized frontend performance for 3+ secure web applications, reducing page load times by 20% using React and Tailwind CSS.
- Architected and implemented responsive UI components for chapter projects, improving user engagement and accessibility for 500+ campus members.
- Coordinated a large-scale technical examination for students, managing logistics and execution to ensure a 100% smooth event delivery.
- Spearheaded web security workshops and awareness campaigns, educating the community on OWASP Top 10 vulnerabilities and secure coding practices.

PROJECTS:

1.LearnX

- Developed an AI-driven educational platform using the Google Gemini API that served 200+ users, improving concept retention by 30% through automated visual knowledge graphs.
- The platform includes dynamic AI-generated quizzes with instant feedback and performance tracking.
- Users can monitor their progress, daily activity, and learning streaks through interactive dashboards.
- Additionally, LearnX supports community sharing, allowing users to publish and clone learning content.

2.Resume Analyzer

- Developed a resume parsing tool using Python and NLP that identified key skill gaps, helping users improve their ATS match rate by an average of 25%.
- Integrated the Natural Language Toolkit (NLTK) to extract and categorize skills, education, and experience from PDF and DOCX formats
- Engineered a keyword-matching algorithm to compare candidate profiles against job requirements, providing automated optimization suggestions

3.Chatbot

- Engineered a neural network-based conversational system using NLP to deliver accurate, context-aware responses in real time.
- Implemented advanced tokenization and sentiment analysis to improve response accuracy for learning and problem-solving tasks.
- Optimized interaction latency, making user assistance 30% faster and more efficient.

4.Ai Tools Tracker

- Automated web scraping and data collection from 3 major tech sources (Product Hunt, Dev.to, Hacker News), reducing manual research time by 10+ hours weekly.
- Developed a centralized database using PostgreSQL to store tool features and categories in a structured format.
- Built a comparison engine that allows users to explore and filter 500+ AI resources based on specific use cases.

5.Smart Irrigation System

- Smart Irrigation System | IoT + AI-Based Automation
- Sensors, Firebase Realtime DB, Embedded Systems
- Built an AI-driven smart irrigation system integrating soil moisture sensors with Firebase, enabling real-time monitoring and reducing water wastage by ~40%
- Automated motor control based on sensor data, eliminating manual intervention and improving irrigation efficiency and reliability
- Designed data-driven decision logic to dynamically adjust irrigation schedules based on environmental conditions

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PUBLICATIONS:

- Filed a patent application with the Government for a Smart Irrigation System featuring AI-based decision making, soil moisture sensors, and automated irrigation control.